

B i G L S G

B i G R U - L S T M a n d
G A N - b a s e d I D S f o r
t h e I o T N e t w o r k

By Sakshi 2021BCS-059

Under the supervision of Dr. Saumya Bhadauria



OUTLINE

- 1 Problem statement
 - 2 Literature Review
 - 3 Proposed Solution
 - 4
 - 5 Workflow of BiGLSG
 - 6 Workflow of GAN
 - 7 Performance matrices
- 

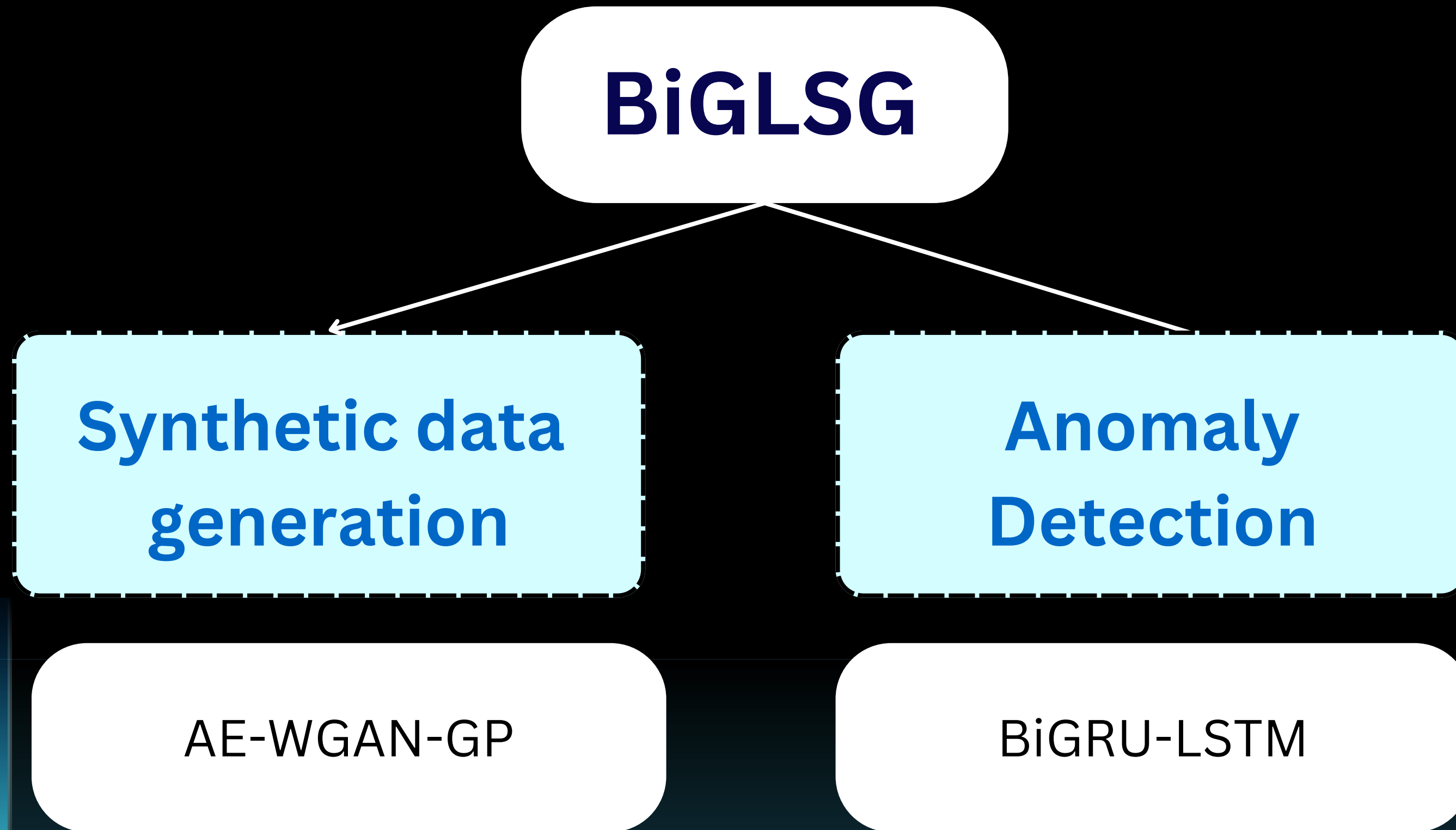
PROBLEM STATEMENT

- The Internet of Things (IoT) refers to the network of physical devices or objects called 'things'.
- These devices are embedded with sensors, actuators, software, etc., which enable them to gather data and share it with other internet-connected devices without human interactions.
- Nowadays, IoT has application in almost every sector, like health, agriculture, and industries, etc.
- The deployment of IoT systems in smart agriculture (SA) operates in extreme environments, including wind, snowfall, landscape, etc.
- IoT devices have various constraints, like **storage constraints**, **computational power constraints**, etc.
- Due to these constraints, IoT devices are vulnerable to various cyberattacks.

PROBLEM STATEMENT CONTD

- An Intrusion Detection System (IDS) can protect IoT devices from various attacks.
- An IDS is a security mechanism that is designed to detect and alert the administrator of any unauthorized or malicious activities in IoT devices or networks.
- Over the past decade, deep learning (DL)-based IDS have gained considerable momentum because of their quick response against anomalies and more meticulous results compared to ML techniques.
- DL-based IDS requires a lot of data to learn complex features; however, the available datasets have class imbalance problems, which makes training of the IDS difficult.
- Generative Adversarial Networks (GAN) are used to resolve the class imbalance problem.

PROBLEM STATEMENT CONTD



LITERATURE REVIEW

Anomaly based IDS

- Since we are handling sequential data, it is necessary to use Recurrent Neural Networks (RNN). However, RNNs suffer from the vanishing gradient problem, which is effectively addressed by Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks.
- Due to its simpler structure and lower computational complexity, GRU is a less complex variant of LSTM.
- LSTM has the potential to learn more intricate patterns and relationships in data. It is particularly well-suited.
- GRU processes data only in the forward direction. To capture dependencies in both forward and backward directions, we use Bidirectional GRU (BiGRU).

LITERATURE REVIEW

GAN

- **Mode Collapse:** The generator produces limited variations, reducing data diversity.
- **Training Instability:** The adversarial nature of GANs makes training difficult and unstable.
- **Vanishing/Exploding Gradients:** The discriminator may become too strong or weak, leading to poor gradient updates.
- **Non-Convergence:** GANs may not reach equilibrium, causing unstable outputs.
- **Evaluation Difficulty:** Assessing GAN performance is challenging due to the lack of a clear loss metric.

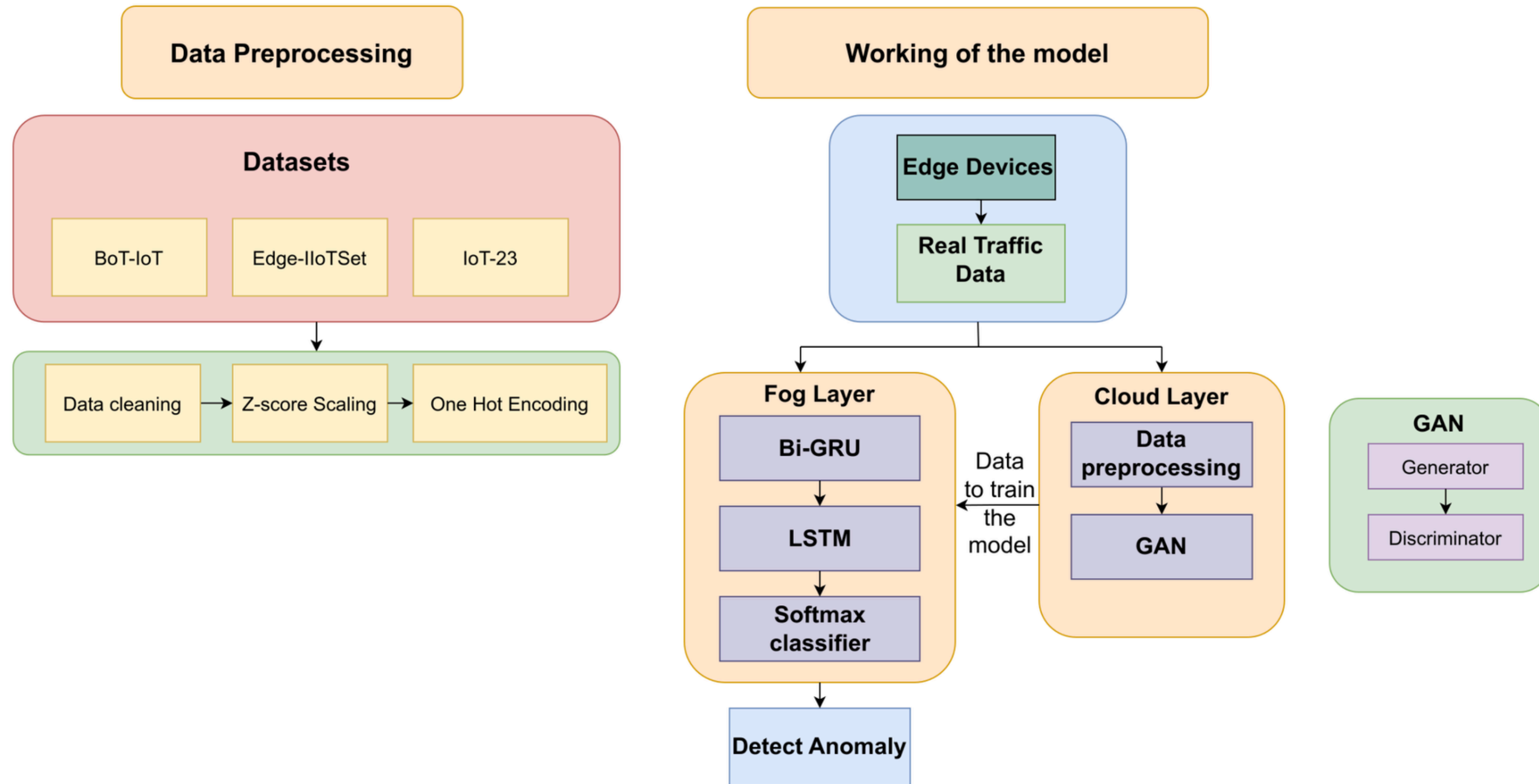
PROPOSED SOLUTION

Design and develop an AE-WGAN-GP to address the problems.

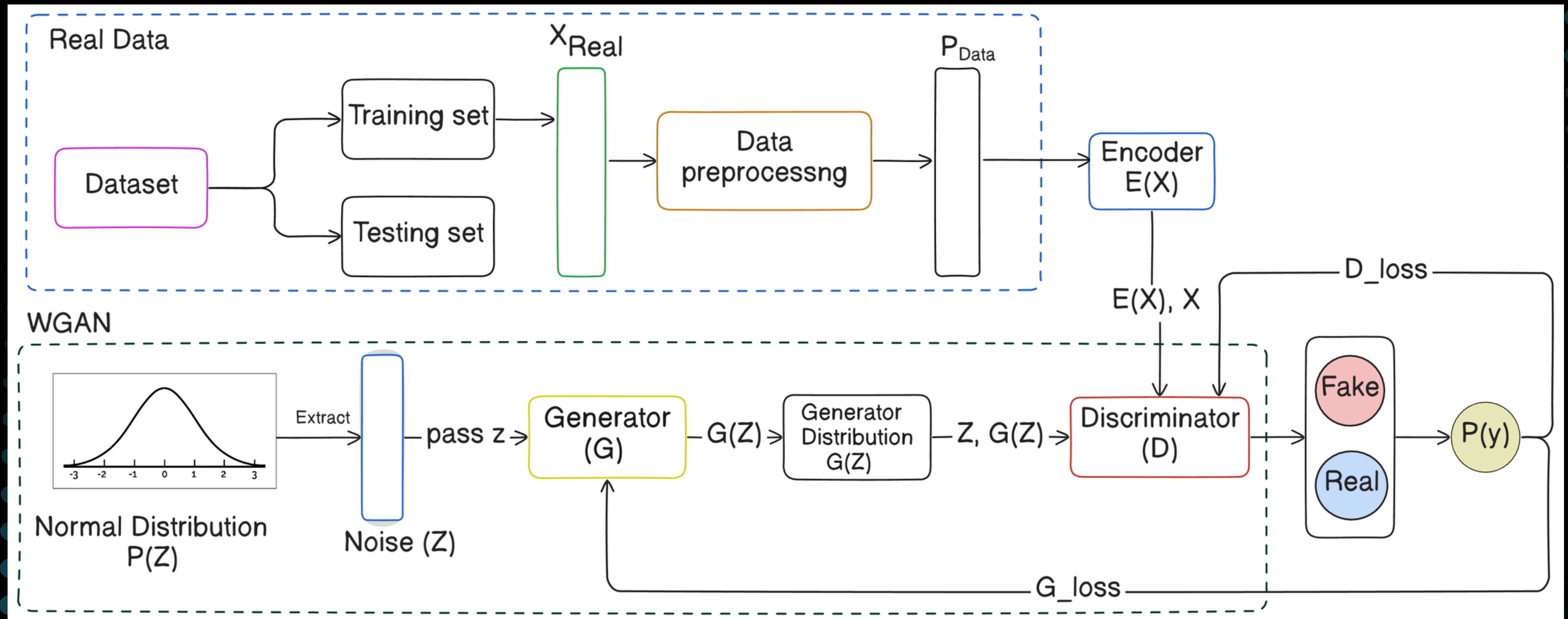
Features of AE-WGAN-GP

- **Autoencoder for Feature Extraction:** An autoencoder for feature extraction that helps in bidirectional learning.
- **Wasserstein Distance:** Ensures stable training by providing a meaningful loss function.
- **Gradient Penalty (GP):** Prevents exploding or vanishing gradients by enforcing smooth gradients.
- **Improved Convergence:** Encourages balanced generator-discriminator learning.
- **Better Sample Quality:** Generates more realistic and diverse samples by learning complex data distributions.

WORKFLOW OF BIGLSC



WORKFLOW OF AE-WAN-GP



Autoencoder Wasserstein GAN with Gradient Penalty

WORKING AE-WAN-GP

Standard GAN Objective function

$$\min_G \max_D V(D, G) = \mathbb{E}_x [\log D(x)] + \mathbb{E}_z [\log(1 - D(G(z)))]$$

AE-GAN objective function

$$V(D, G, E) = \mathbb{E}_x \mathbb{E}_z p(z|x) [\log(D(\{x, E(x)\}; \theta_D))] \\ + \mathbb{E}_z \mathbb{E}_x p(x|z) [\log(1 - D(\{G(z; \theta_G), z\}; \theta_D))]$$

AE-WGAN-GP

$$L_{\text{Critic}} = \mathbb{E}_x [D(x)] - \mathbb{E}_x [D(G(x))] + GP$$

$$GP = \lambda \cdot \mathbb{E}_{\hat{x}} [||\nabla_{\hat{x}} D(\hat{x})|| - 1]^2$$

PERFORMANCE MATRICS

The following evaluation metrics can be used for the evaluation of IDS.

- **True Positive (TP):** The data instances that are correctly classified as an attack instance.
- **True negative (TN):** The data instances that are correctly classified as normal instances.
- **False Negative (FN):** The data instance that is wrongly classified as a normal instance.
- **False Positive (FP):** The data instance that is wrongly classified as an attack instance.
- **Accuracy:** It is the ratio of correctly classified instances to the total no of instances in a dataset.

PERFORMANCE MATRICS CONTD

- **Precision:** It is the ratio of correctly predicted instances as an attack to all the instances predicted as an attack.
- **Recall:** It is the ratio of all correctly classified instances as attacks to all instances that really attack.
- **F1-Score:** The F1 score is an evaluation metric that measures a model's accuracy. It is a weighted average of precision and recall.
- **ROC (Receiver Operating Characteristic):** The data instance that is wrongly classified as a normal instance.

CONCLUSION

- BiGRU-LSTM for Anomaly Detection: Efficiently captures sequential patterns in IoT network traffic.
- AE-WGAN-GP for Synthetic Data Generation: Balances class distribution by generating realistic attack samples.
- Addresses Class Imbalance: Helps the IDS learn from both majority and minority classes effectively.
- Wasserstein Distance: Stabilizes GAN training by providing a meaningful loss function.
- Gradient Penalty (GP): Prevents mode collapse and ensures smooth gradient updates.
- Enhanced IDS Performance: Improves detection accuracy, reduces false positives, and increases robustness against cyber threats.

The background is a dark navy blue. On the left side, there are several overlapping, semi-transparent blue geometric shapes, including rectangles and parallelograms, some of which are tilted. On the right side, there is a grid of small, light blue dots that fade out towards the center.

THANK YOU